

Aadarsh Katigi

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EDUCATION

University of Massachusetts Boston

Boston, MA

Master of Science | Computer Science

Sep 2023 - May 2025

Relevant Coursework: Applied Machine Learning, AI, Neural Networks, Distributed Systems, Operating Systems, OOP

Visvesvaraya Technological University

Belgaum, India

Bachelor of Engineering | Computer Science

Aug 2017 - Aug 2021

WORK EXPERIENCE

Bank of New York

Austin, TX

Software Engineer

Jan 2025 - Present

- Architected **45+ REST APIs** for a financial trading platform handling **\$2T+** in assets with **99.99%** availability, by orchestrating **Java Spring Boot microservices** on **Kubernetes**.
- Eliminated read-latency bottlenecks for pipelines ingesting **10TB+** monthly logs, achieving **sub-50ms** reads, by implementing write-behind caching with **Redis** and **PostgreSQL read replicas**.
- Secured platform access for **100K+ users** and reduced login failures by **28%**, by engineering a centralized authentication service using **Node.js**, **OAuth**, and **JWT**.
- Enhanced code quality, increasing **JUnit** unit test coverage from **65%** to **82%** while reducing post-deployment bugs by **22%** quarterly, by using **GitHub Copilot** to accelerate test development.

Tata Consultancy Services

Bengaluru, India

Software Engineer

Sep 2021 - Jul 2023

- Architected scalable, reliable **Java** software solutions supporting **200K+** global servers and processing **2M+** daily messages, by designing decoupled **Kafka** consumers with object-oriented design patterns in **Spring Boot**.
- Diagnosed cascading failures across distributed systems, saving **15 hours** of weekly on-call time, by refactoring routing logic with the **Strategy Pattern** and exponential backoff.
- Owned code deployment pipelines and dashboards for operational excellence, reducing monthly **AWS** spend by **\$20K**, by rightsizing **EC2** instances and configuring auto-scaling.
- Delivered high-quality software on tight schedules by conducting code reviews, reducing production rollbacks by **15%**, by integrating **JUnit** and **Mockito** test suites into build processes.

ATIL LLP

Belgaum, India

Software Engineer Intern

Feb 2020 - Jun 2020

- Designed a highly performant **C++17** backend serving **200K+** concurrent requests with **sub-10ms** latency, by implementing architecture with a custom thread pool and the **Reactor pattern** using **epoll**.
- Architected a fault-tolerant distributed queue processing **2M+** daily transactions with **zero data loss**, by building reliable **C++** routines and utilities with **ZeroMQ** and automated failover.
- Engineered a database caching layer reducing read latency by **90%** for frequently accessed records, by developing an **LRU cache** to shield the **MySQL** database from high-throughput loads.

PROJECTS

Production Incident Copilot | Python, TensorFlow, scikit-learn, LangChain

Austin, TX

- Built an ML classifier to detect recurring failure signatures in production systems, cutting false-negatives by **20%**, by training models with **TensorFlow** and **scikit-learn** on historical data.
- Integrated retrieval-driven RCA into the incident workflow, outputting actionable remediation steps, by leveraging **LangChain** to fetch similar incidents and runbooks.

TECHNICAL SKILLS

Languages: C++, Java, Python, SQL, TypeScript, JavaScript, C, Shell Scripting

Frameworks: Spring Boot, Hibernate, Spring Security, Django, FastAPI, Kafka, React, ReactNative, Next.js, LangChain, TensorFlow

Databases: PostgreSQL, Redis, Elasticsearch, MySQL, DynamoDB, Cassandra, Oracle DB

Developer Tools: AWS (EC2, S3, CloudWatch), GCP, Kubernetes, Docker, Terraform, Git, GitHub Actions, JUnit, Mockito

Core Competencies: C++ & Java Architecture, Object-Oriented Design, Design Patterns, High-Performance Systems, Operational Excellence, Monitoring & Dashboards, Full SDLC, Code Reviews, Build Processes, Reliability & Scaling, CI/CD